Assignment 3: Database Design Analysis

Database: MySQL | Project: Airline Reservation System

1. Objective

Analyze the Airline Reservation System design for data redundancy, insertion, deletion and update anomalies, functional dependencies, and normalization improvements.

2. Introduction

The Airline Reservation System manages passengers, aircraft, flights, bookings, and payments. Storing all information in one table would repeat passenger and flight details and create maintenance problems. A normalized design separates the entities and connects them through keys.

3. Example of redundancy

Passenger ID	Passenger Name	Flight	Aircraft	Booking
101	Arjun Sharma	6E101	Airbus A320	B001
101	Arjun Sharma	6E404	Airbus A320	B002
102	Priya Nair	6E101	Airbus A320	B003
103	Rahul Kumar	AI202	Airbus A321	B004

Passenger and aircraft information can be repeated whenever the same passenger or aircraft appears in several booking records.

4. Anomalies

Insertion anomaly: A new aircraft or flight may be difficult to store if the design requires a booking before the related details can be recorded.

Update anomaly: Changing aircraft capacity, passenger contact details, or flight information in several repeated rows can produce inconsistent values if one row is missed.

Deletion anomaly: Removing the only booking for a flight could accidentally remove the only stored information about that flight or its aircraft.

5. Functional dependencies

Table / Entity	Functional dependency
Passengers	passenger_id → name, email, phone, passport_no
Aircraft	aircraft_id → model, capacity, airline_name
Flights	flight_id → flight_number, departure_airport, arrival_airport, departure_time, arrival_time, aircraft_id
Bookings	booking_id → booking_date, seat_number, class, passenger_id, flight_id
Payments	payment_id → payment_date, amount, payment_method, payment_status, booking_id

The UNIQUE attributes email, passport_no and flight_number can also identify records uniquely within their respective tables.

6. Normalization

1NF: Each field contains one atomic value. Passenger, flight, seat and payment details are not stored as repeating lists in a single row.

2NF: Non-key attributes depend on the complete key. Booking information is associated with a specific booking, passenger and flight relationship.

3NF: Non-key attributes depend on the key rather than on other non-key attributes. Aircraft details are stored once in Aircraft, while passenger details are stored once in Passengers.

7. Improvements

Create separate entity tables; assign primary keys; use foreign keys; apply UNIQUE, NOT NULL, CHECK and DEFAULT constraints; use a dedicated Booking table to resolve the many-to-many passenger-flight relationship; and verify valid and invalid operations.

8. Conclusion

Normalization reduces duplication, prevents insertion, deletion and update anomalies, and improves the consistency and maintainability of the Airline Reservation System.